

District 5 Model Comparison - Geospatial Data: 2023 - 2025 Sales

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21 July 2026

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Model I

Table 1: Model Metrics: LN-SPPSF Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	161	153	7	0.6267	0.3928	0.3650	0.6275	14.1386	316	344
+ Geospatial	161	151	9	0.6630	0.4396	0.4062	0.6068	13.1595	307	341
% Difference	0.00%	-1.31%	28.57%	5.79%	11.91%	11.28%	-3.30%	-6.93%	-2.85%	-0.87%

Table 2: Ratio Metrics: LN-SPPSF Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9132	1.2227	0.8670	1.4103	62.8191	-0.5750	14.29%	29.81%	59.01%
+ Geospatial	0.9451	1.2078	0.8770	1.3773	58.6912	-0.4787	10.56%	25.47%	65.22%
% Difference	3.49%	-1.21%	1.15%	-2.34%	-6.57%	16.74%	-26.09%	-14.58%	10.53%

Model II

Table 3: Model Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	308	289	18	0.7312	0.5346	0.5057	0.6744	18.4465	651	726
+ Geospatial	308	289	18	0.7410	0.5491	0.5210	0.6638	19.5489	642	716
% Difference	0.00%	0.00%	0.00%	1.34%	2.69%	3.03%	-1.56%	5.98%	-1.38%	-1.38%

Table 4: Ratio Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9614	1.2379	0.8589	1.4412	62.7751	-0.3361	13.96%	25.65%	58.12%
+ Geospatial	0.9666	1.2310	0.8617	1.4285	62.0031	-0.3214	12.66%	22.73%	59.42%
% Difference	0.54%	-0.56%	0.33%	-0.88%	-1.23%	4.35%	-9.30%	-11.39%	2.23%

Model III

Table 5: Model Metrics: LN-PRICE Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	308	281	26	0.8050	0.6480	0.6154	0.5948	19.8957	581	686
+ Geospatial	308	282	25	0.8000	0.6400	0.6081	0.6005	20.0537	586	687
% Difference	0.00%	0.36%	-3.85%	-0.62%	-1.23%	-1.19%	0.95%	0.79%	0.86%	0.15%

Table 6: Ratio Metrics: LN-PRICE Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9803	1.1826	0.9003	1.3136	51.5571	-0.2192	17.53%	28.90%	64.61%
+ Geospatial	0.9662	1.1878	0.8958	1.3260	53.4775	-0.2256	14.61%	30.19%	63.64%
% Difference	-1.43%	0.44%	-0.50%	0.95%	3.72%	-2.89%	-16.67%	4.49%	-1.51%

Model IV

Table 7: Model Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	261	237	23	0.8813	0.7767	0.7550	0.4452	35.8318	343	432
+ Geospatial	263	240	22	0.8803	0.7750	0.7544	0.4496	37.5756	349	435
% Difference	0.77%	1.27%	-4.35%	-0.11%	-0.21%	-0.08%	0.98%	4.87%	1.75%	0.69%

Table 8: Ratio Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	1	1.0948	0.9384	1.1667	36.3029	-0.1106	18.01%	36.02%	74.33%
+ Geospatial	1	1.0974	0.9398	1.1677	36.8862	-0.1123	19.77%	35.74%	74.90%
% Difference	0.00%	0.24%	0.15%	0.09%	1.61%	-1.54%	9.80%	-0.76%	0.77%

Model V

Table 9: Model Metrics: LN-TASP Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	261	239	21	0.8758	0.7671	0.7466	0.4473	37.4852	343	425
+ Geospatial	263	241	21	0.8736	0.7631	0.7424	0.4522	36.9654	351	434
% Difference	0.77%	0.84%	0.00%	-0.26%	-0.52%	-0.56%	1.10%	-1.39%	2.33%	2.12%

Table 10: Ratio Metrics: LN-TASP Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	1.0054	1.0966	0.9330	1.1754	36.4828	-0.1172	20.31%	36.78%	74.71%
+ Geospatial	0.9926	1.0989	0.9328	1.1781	37.4377	-0.1224	18.25%	34.98%	74.52%
% Difference	-1.28%	0.21%	-0.03%	0.23%	2.62%	-4.48%	-10.12%	-4.90%	-0.25%

Model VI

Table 11: Model Metrics: LN-TASP Model II Comparison

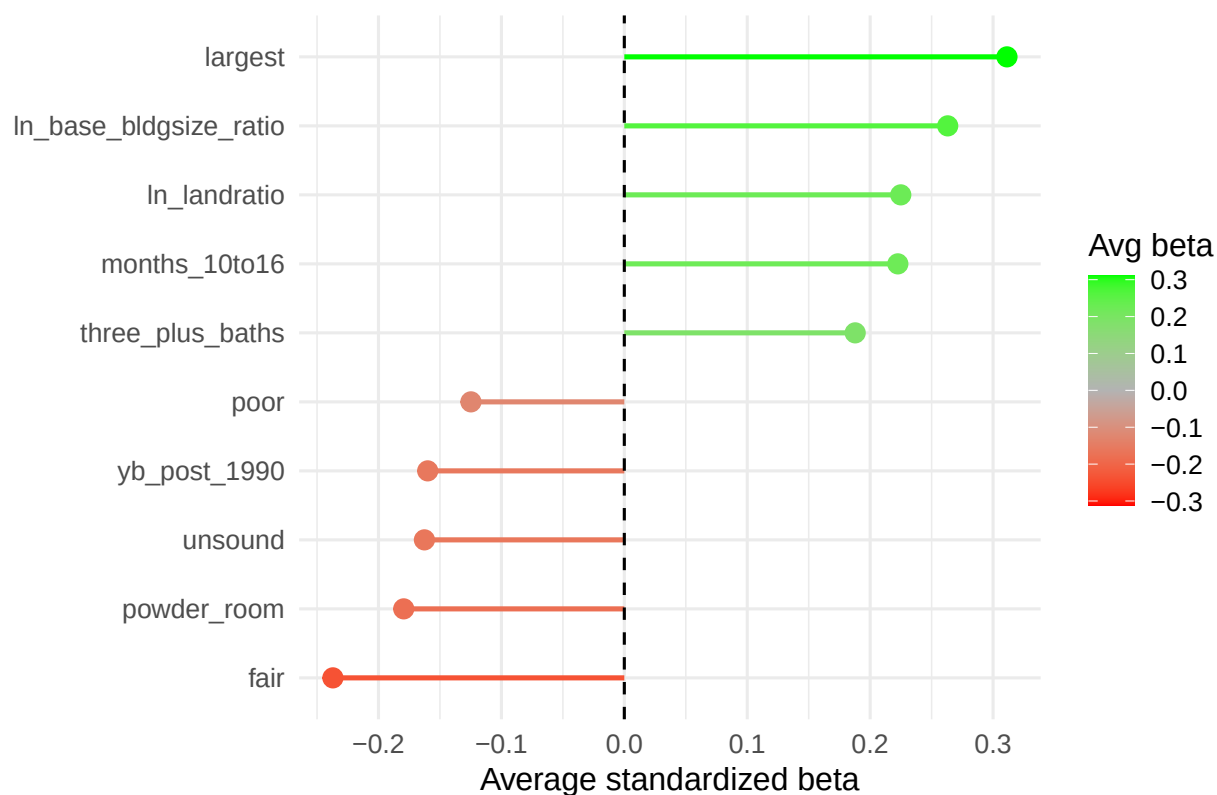
Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	261	239	21	0.8758	0.7671	0.7466	0.4473	37.4852	343	425
+ Geospatial	263	243	19	0.8708	0.7582	0.7393	0.4550	40.1066	353	428
% Difference	0.77%	1.67%	-9.52%	-0.58%	-1.16%	-0.98%	1.71%	6.99%	2.92%	0.71%

Table 12: Ratio Metrics: LN-TASP Model II Comparison

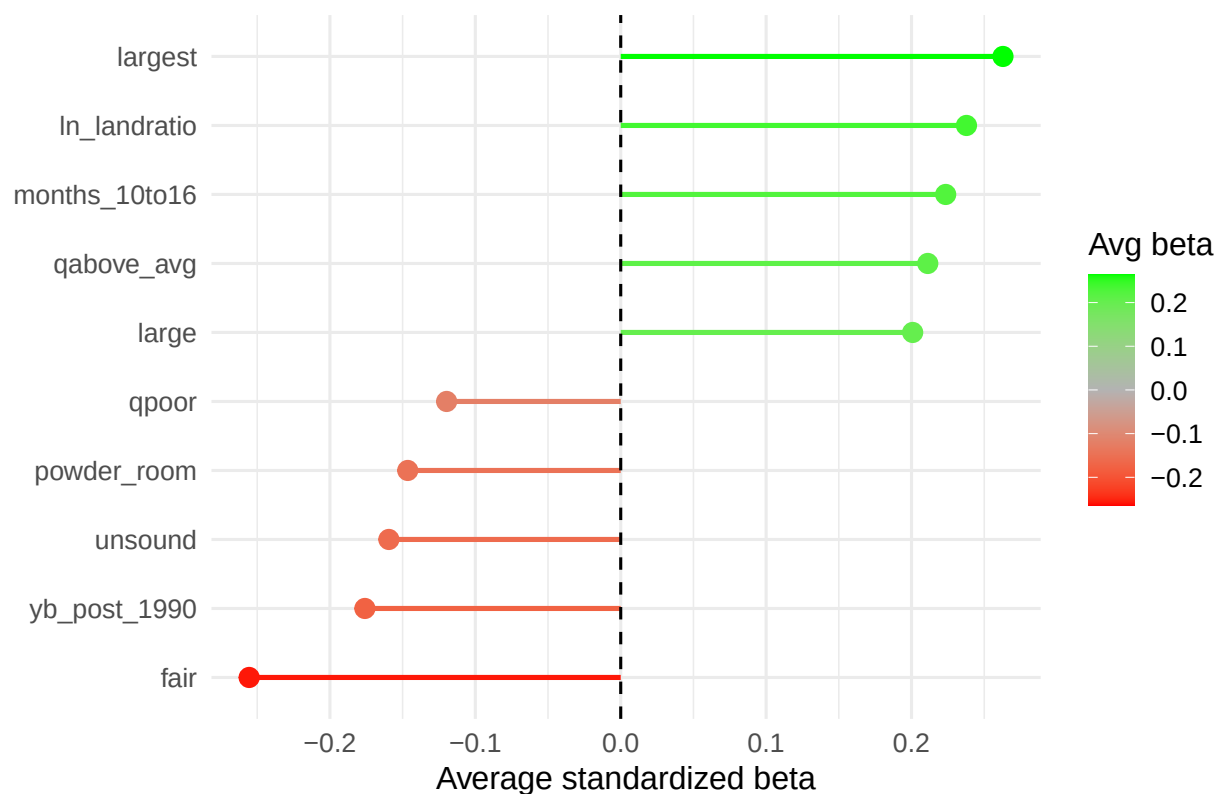
Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	1.0054	1.0966	0.9330	1.1754	36.4828	-0.1172	20.31%	36.78%	74.71%
+ Geospatial	0.9865	1.1012	0.9328	1.1805	38.3120	-0.1239	15.97%	35.36%	74.90%
% Difference	-1.88%	0.42%	-0.02%	0.44%	5.01%	-5.79%	-21.36%	-3.86%	0.26%

Model Comparison

District 5 | Avg Std Beta across Initial Models



District 5 | Avg Std Beta across Geospatial Models



Prediction Testing

Table 13: Deviation Summary: Initial LN-SPPSF Model

Metric	Value
Average sale price deviation	-802.19
Max deviation	449069.71
Min deviation	-1097629.62
Standard deviation	122883.95

Table 14: Deviation Summary: LN-SPPSF Model + Geospatial

Metric	Value
Average sale price deviation	4309.45
Max deviation	469482.60
Min deviation	-1088424.34
Standard deviation	122677.95